

PHYSICS: KINEMATIC MOTION & GRAPHING CHALLENGE

UNIT 2: CONSTANT SPEED, COORDINATE PLOTTING • CHALLENGE 1 OF 2

MISSION SCENARIO 1: HARBOR COAST GUARD AIR-DROP

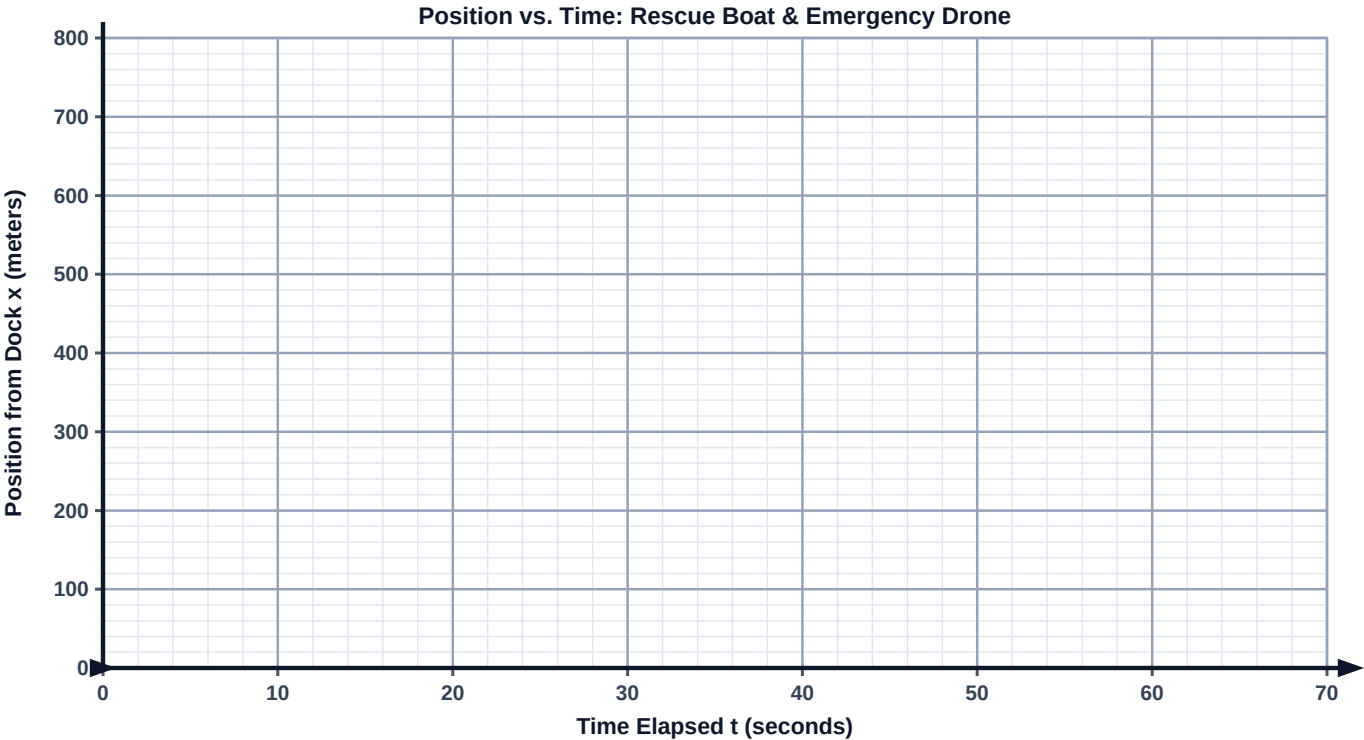
A rescue boat leaves the harbor dock ($x = 0\text{ m}$) and cruises straight out to sea at a steady speed of 6 m/s . At $t = 0\text{ s}$, the boat is already **300 meters** out.

A fast emergency drone is launched from the dock ($x = 0\text{ m}$). The drone must wait **20 seconds** for GPS lock, remaining on the dock ($x = 0\text{ m}$ from $t = 0$ to 20 s), and then flies at a fast steady speed of **18 m/s** to deliver supplies.

Rescue Boat (Steady Speed = 6 m/s) Starts at 300 m at $t = 0\text{ s}$ (+60 m every 10 s):							
Time (s)	0	10	20	30	40	50	60
Pos (m)	300	360	420	480	540	600	660

Supply Drone (Delayed Takeoff, Speed = 18 m/s) Waits on dock (0 m) until $t = 20\text{ s}$, then adds +180 m every 10 s:							
Time (s)	0	20	30	40	50	60	
Pos (m)	0	0	180	360	540	720	

Graphing Step: Plot the coordinates from the tables above. Draw straight lines through the points. Circle the point where the two lines cross (meet)! VISUAL GRAPH ANALYSIS



GRAPH ANALYSIS QUESTIONS USE YOUR GRAPH ABOVE TO FIND ANSWERS

- 1. The Rendezvous Point (Where Lines Cross):**
Look at where the Boat line and Drone line intersect on your graph:

 - At what **Time** does the drone catch up to the boat? _____ seconds
 - What is their **Position** (distance from dock) at that moment? _____ meters
- 2. Comparing Steepness & Speed:**
Which line is steeper on the graph (Boat or Drone)? _____ • In physics, what does a **steeper slope** tell you about an object's speed?
- 3. Actual Flying Time:**
The clock reads your meeting time when they meet, but the drone was waiting on the dock for the first 20 seconds. How many seconds was the drone actually flying in the air? (Meeting Time – 20 s) = _____ seconds
- 4. Battery Range Check:**
The drone battery allows a total round-trip distance of **1,200 meters** (to the boat and back to the dock). If the drone flies out to the meeting position and immediately flies back, what is its total round-trip distance ($2 \times$ position)? Can it make it back safely? Explain:

PHYSICS: HIGHWAY MOTION & INTERCEPT CHALLENGE

Name: _____ Period: _____ Date: _____

UNIT 2: MOVING, STOPPING & SPEED CALCULATIONS • CHALLENGE 2 OF 2

MISSION SCENARIO 2: HIGHWAY COURIER JOURNEY & PATROL INTERCEPT

A delivery truck drives along a straight highway from the station ($x = 0\text{ m}$) in three distinct legs:

- **Leg 1 (0 to 20 s):** Drives forward at 15 m/s → Reaches $x = 300\text{ m}$ at $t = 20\text{ s}$.
- **Leg 2 (20 to 50 s):** Stops at a rest area (speed = 0 m/s) → Remains at $x = 300\text{ m}$ for 30 seconds.
- **Leg 3 (50 to 90 s):** Resumes driving at 25 m/s → Reaches $x = 1,300\text{ m}$ at $t = 90\text{ s}$.

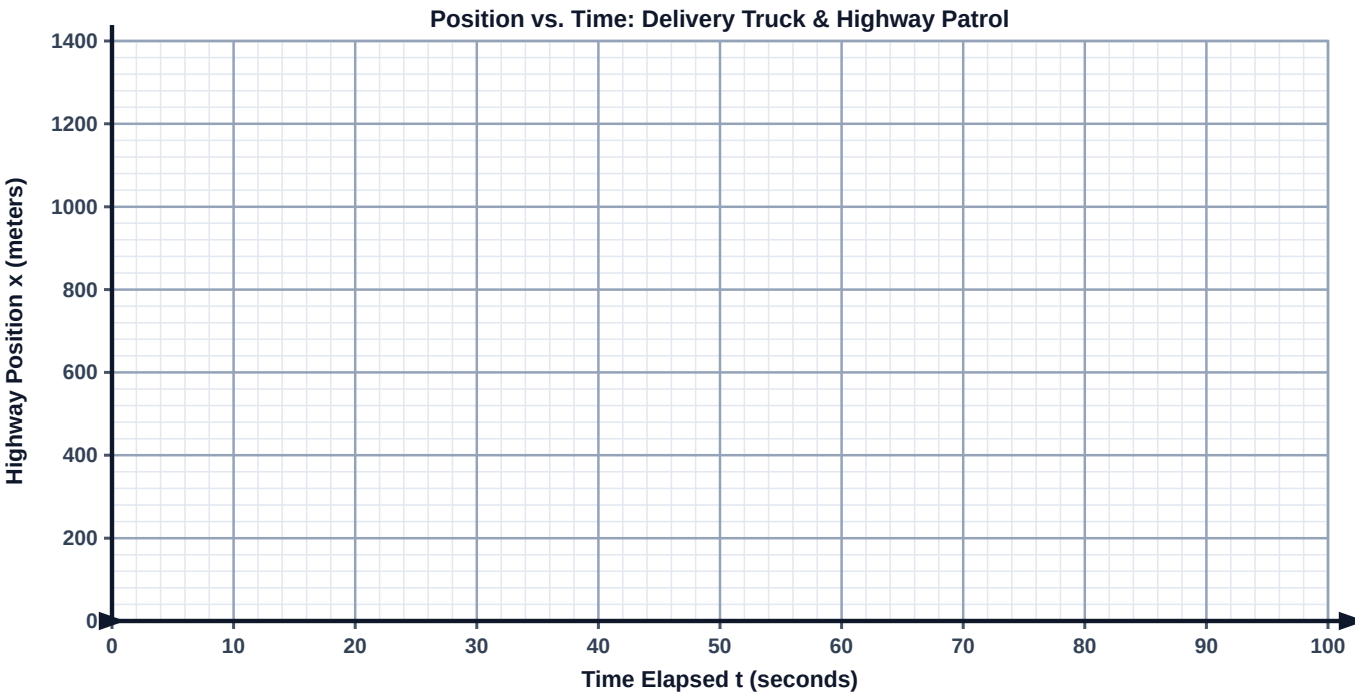
Delivery Truck Coordinates
Plot these 4 checkpoints and connect with lines:

Time (s)	0	20	50	90
Pos (m)	0	300	300	1,300
Status	Start	Rest In	Rest Out	Finish

Highway Patrol Intercept Lines
Leaves station at $t = 30\text{ s}$ ($x = 0\text{ m}$):

Target	Start Coordinate	Target Intercept
Line A	(30 s, 0 m)	(50 s, 300 m) • Rest Stop
Line B	(30 s, 0 m)	(90 s, 1,300 m) • Highway End

Graphing Step: 1) Plot the truck's 3-part journey. 2) Draw Patrol Line A (to rest stop). 3) Draw Patrol Line B (to end). FLAT LINE = STOPPED



GRAPH ANALYSIS QUESTIONS USE YOUR GRAPH AND FORMULAS TO ANSWER

- 1. Understanding the Rest Stop (Flat Line):**
What does the completely flat horizontal line between $t = 20\text{ s}$ and $t = 50\text{ s}$ tell you about the truck's motion? How do you know the truck is NOT moving?
- 2. Calculating Patrol Car Speeds (Speed = Distance / Time):**
- **Option A (Catch at rest stop):** Car travels 300 m in 20 seconds (from 30 s to 50 s).
Speed = $300\text{ m} \div 20\text{ s} = \underline{\hspace{2cm}}\text{ m/s}$
 - **Option B (Catch at end):** Car travels 1,300 m in 60 seconds (from 30 s to 90 s).
Speed = $1,300\text{ m} \div 60\text{ s} = \underline{\hspace{2cm}}\text{ m/s}$
- 3. Patrol Safe Speed Limit:**
The patrol car has a maximum safe speed limit of **20 m/s** (about 45 mph). Which option can the patrol car safely choose (Option A or Option B)? Explain why using your speeds calculated above:
- _____
- _____